MURUGANANDAM M G

AI Developer | Python Developer | GenAI & ML Systems

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate and CTO at A-Zentrix, building Python, FastAPI and React applications. Projects cover Git-native coding automation, LangChain-powered productivity tools, coronary-risk ML with 3D visualization, and computer vision. I focus on implementation: APIs, model evaluation, testing and maintainable code.

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, SQL

AI & GenAI: Machine Learning, Deep Learning, LLMs, RAG, AI Agents, NLP, Computer Vision, Prompt Engineering

Frameworks & Libraries: FastAPI, Flask, LangChain, LangGraph, scikit-learn, CatBoost, XGBoost, OpenCV

Web & Databases: React (TypeScript), HTML, CSS, Tailwind CSS, PostgreSQL, MySQL, SQLite, ChromaDB

Tools & DevOps: Git, GitHub, Docker, Ollama, VS Code, Cursor, pytest, Kubernetes (basic)

Core Concepts: REST APIs, Backend Development, Model Evaluation, Vector Search, Tool Orchestration, Debugging

EXPERIENCE

Chief Technology Officer (CTO) — A-Zentrix | Current role

- Work on technical direction and application architecture for A-Zentrix projects, linking AI components, backend services and web interfaces.
- Contribute to the company's React/TypeScript web application; use Git and GitHub to manage source code and coordinate development.

Team Lead — TuneEng | 2025–2026

- Led the TuneEng project team developing an AI-assisted English communication platform with speaking practice, language feedback and interview-preparation features.
- Coordinated development tasks and integration work across frontend, backend and AI-driven learning features, supporting team communication and project progress.

AI & Full-Stack Project Developer — Student Projects | 2023–Present

- Build AI prototypes and supporting applications with Python, FastAPI and React; work across model integration, API design, debugging and evaluation.

SELECTED PROJECTS

Agent Forge — Git-Native Coding Agent

A terminal-based coding assistant that plans, edits and validates changes inside a Git repository.

- Designed a workflow for understanding the working tree, gathering relevant code context and breaking a development request into executable tasks.
- Implemented Git-oriented patch handling and sandboxed evaluation so proposed file changes can be checked before they affect the primary workspace.
- Connected LLM-assisted code changes with pytest validation and iterative repair workflows, emphasizing traceable, testable modifications.

Tech Stack: Python | Git | LLM APIs | pytest | CLI | Sandbox Execution

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Aura AI — LangChain LLM Workflows

A habit, focus and personal-growth application that combines structured activity tracking with an AI coaching interface.

- Developed FastAPI endpoints and LangChain prompt workflows to support conversational mentoring using Gemini and OpenAI model integrations.
- Implemented authenticated user workflows for routines, progress check-ins and focus sessions using asynchronous SQLAlchemy and SQLite.
- Connected habit analytics, focus-session logging, AI guidance and calendar/CSV exports to a web interface, with automated tests documented in the repository.

Tech Stack: Python | FastAPI | LangChain | Gemini | OpenAI | SQLAlchemy | SQLite

CardioVision AI — IIT Mandi Hackathon

A cardiovascular decision-support prototype that combines coronary artery risk prediction with interactive 3D anatomical interpretation.

- Built a multi-target ML workflow on the UCI Z-Alizadeh Sani coronary dataset to estimate coronary artery disease and LAD, LCX and RCA vessel-specific stenosis risk.
- Linked risk outputs to a Three.js heart visualization that lets users inspect coronary regions and their predicted risk levels spatially.
- Included explainability and evaluation views, such as SHAP-based feature contributions, ROC/precision-recall curves, threshold analysis and evidence reports.

Tech Stack: Python | FastAPI | LightGBM | React | TypeScript | Three.js | SHAP

Neural Age & Gender Engine — Computer Vision

A Flask-powered image analysis prototype with face localization and age/gender-estimation outputs.

- Created a Python/Flask image-processing flow that accepts uploaded images, processes faces using OpenCV and returns structured results for the interface.
- Built a responsive browser-based upload screen with face bounding-box overlays, confidence display and adjustable inference delay for workflow testing.

Tech Stack: Python | Flask | OpenCV | NumPy | HTML | CSS | JavaScript

EDUCATION

B.Tech — Artificial Intelligence & Data Science | 2024–2028 (Expected)
Nandha Engineering College (Autonomous), Perundurai, Erode, Tamil Nadu

CERTIFICATIONS

LinkedIn Learning — Advanced Prompt Engineering Techniques
Great Learning — Introduction to Neural Networks & Deep Learning
SWAYAM (Skill India) — CLOUD COMPUTING
IBM Cognitive Class. — Machine Learning with Python

PROFESSIONAL MEMBERSHIPS

IEEE — Member